

The big picture

Guiding question(s)

What role do bonding and structure have in the design of materials?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

The field of materials science is an interdisciplinary field, often combining chemistry, physics and engineering to design and discover new materials, typically solids. Materials chemists study how the synthesis of a material affects the resulting structure and thus the properties. The scope of materials science is often depicted as a tetrahedron to highlight the four main interdependent aspects. A version of this tetrahedron is shown in **Figure 1**.

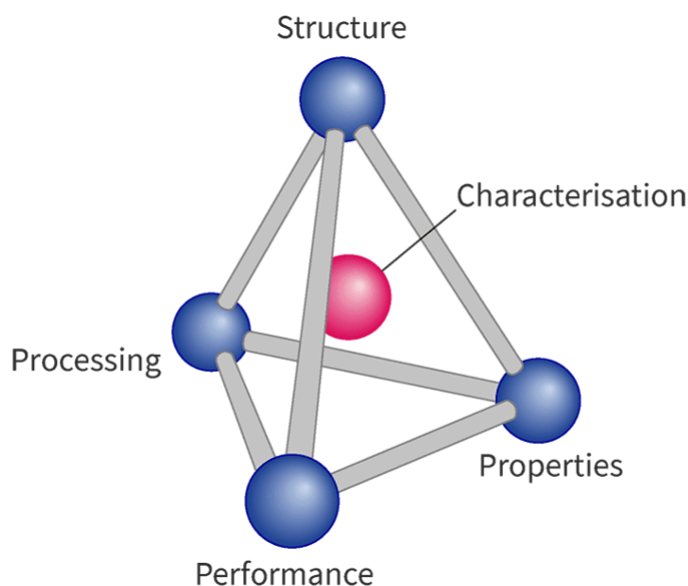


Figure 1. A materials science tetrahedron.

👁 More information for figure 1

You might notice that in the centre of the tetrahedron is ‘characterisation’, this term refers to the qualitative and quantitative measurement of each of the other four points. ‘Performance’ is a measure of how good a material is at achieving its intended purpose and includes scenarios where the material no longer performs as required. ‘Properties’ are macroscopic things that can be measured, we have considered these in [section S2.1.3b \(/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/\)](#) for ionic compounds, [section S2.2.7 \(/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/\)](#) for covalent network substances, and [section S2.3.1b \(/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/\)](#) for metals. ‘Structure’ refers to the structural arrangement of atoms, ions and electrons at the atomic scale but also microscopic and macroscopic structures. ‘Processing’ is the individual steps needed to create a material, including both the synthesis and treatment after synthesis.

The design of most materials involves consideration of the specific properties that are desired. Look around you, what material objects do you see? What specific purpose is that material object used for? The properties of a material are determined by the structure and bonding of its component parts.

Interactive 1 shows different examples of everyday objects. The properties needed for each material to be useful is labelled for each object. As you scroll and look through each of the objects, consider what type of bonding and structure each material might have.



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Interactive 1. Different Examples of Everyday Objects.

 More information for interactive 1

Prior learning

Before you study this subtopic make sure that you understand the:

- structure and bonding of ionic [section S2.1.3a](#)  (/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/), covalent network [section S2.2.7](#)  (/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) and metallic [section S2.3.1a](#)  (/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/) substances
- common properties of ionic [section S2.1.3b](#)  (/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/), molecular solids [section S2.2.9](#)  (/study/app/preview-p/sid-426-cid-753301/book/strength-of-intermolecular-forces-id-43852/), covalent network structures [section S2.2.7](#)  (/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) and metallic [section S2.3.1b](#)  (/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/) substances
- definition of electronegativity and how it can be used to deduce bond polarity [section S2.2.5](#)  (/study/app/preview-p/sid-426-cid-

[753301/book/bond-polarity-id-43853/](#).